

Employment

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| Graduate Research Assistant                                                                                                                                                                                                                                                                                                                                                                                                                                                  | NC State University  | April 2025 – Present |
| <ul style="list-style-type: none"><li>Designed cloud-based genomic platform with JBrowse 2, enabling 100+ researchers to explore 500GB datasets worldwide, deployed on Google Cloud Run.</li><li>Collaborated on ML architecture design, applying hyperparameter tuning and time-series analysis, improving model accuracy by 12%.</li></ul>                                                                                                                                 |                      |                      |
| Product Engineer Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Isomer AI            | May 2025 – Present   |
| <ul style="list-style-type: none"><li>Owned full backend of production AI platform: Temporal.io workflows, event pipelines, PostgreSQL, Redis, Docker, Terraform, Auth0, Sentry; shipped 10+ features from DB schema to API across 7 deployments in TypeScript.</li><li>Engineered LLM pipelines across Claude, GPT-4o, and Gemini via MCP with provider-agnostic interfaces; built document processing (PDF gen, email parsing, S3) and Playwright test coverage.</li></ul> |                      |                      |
| Software Developer Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Mphasis (HP Company) | Jun 2024 – Aug 2024  |
| <ul style="list-style-type: none"><li>Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML; improved claim accuracy by 25%.</li><li>Automated deployments on AWS &amp; GCP, integrated monitoring dashboards with Grafana, and supported model validation and red-teaming to enhance reliability.</li></ul>                                                                                                                          |                      |                      |
| AI/ML Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Capgemini            | Jun 2023 – Aug 2023  |
| <ul style="list-style-type: none"><li>Led end-to-end SDLC for a data-driven system automating support ticket analysis, reducing processing time by 15%.</li><li>Built distributed pipelines with Hadoop + Spark for batch processing and Kafka streaming for real-time ingestion, cutting latency by 20%.</li></ul>                                                                                                                                                          |                      |                      |

Education

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| Raleigh, NC                                                                                                                                                                                                                     | North Carolina State University | Aug 2024 – May 2026 |
| <ul style="list-style-type: none"><li>Master of Science in Computer Science</li><li>Courses: Artificial Intelligence, Operating Systems, Data Science, Object Oriented Development, Software Engineering, Algorithms.</li></ul> |                                 |                     |
| Mumbai, India                                                                                                                                                                                                                   | Mumbai University               | Aug 2020 – May 2024 |
| <ul style="list-style-type: none"><li>Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS &amp; Algorithms, Machine Learning, Blockchain.</li></ul>                        |                                 |                     |

Projects

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| LLM Eval Harness                                                                                                                                                                                                                                                                                        | Independent Project    | 2025                |
| <ul style="list-style-type: none"><li>Built vendor-neutral LLM benchmark comparing Claude, GPT-4o, and Gemini across 2,000 examples and 3 task types.</li><li>Tracked hallucination rate, p50/p95 latency, cost-per-call; applied bootstrap + McNemar tests across 6 metrics.</li></ul>                 |                        |                     |
| PulseRank                                                                                                                                                                                                                                                                                               | Real-Time Leaderboard  | 2026                |
| <ul style="list-style-type: none"><li>Built real-time leaderboard with MySQL outbox, MongoDB TTL projections, and Redis sorted sets for sub-ms reads.</li><li>Shipped OpenTelemetry traces + Prometheus metrics; load-tested with k6 and deployed via GitHub Actions CI/CD.</li></ul>                   |                        |                     |
| VectorLift                                                                                                                                                                                                                                                                                              | Semantic Search Engine | 2026                |
| <ul style="list-style-type: none"><li>Evaluated BM25 (Elasticsearch), dense bi-encoder with in-batch negatives, and cross-encoder re-ranker on MS MARCO.</li><li>Scored NDCG@10, MRR@10, MAP; served via FastAPI with per-stage latency; Kafka-driven corpus indexing into Qdrant.</li></ul>            |                        |                     |
| Demand Paging – Xinu OS                                                                                                                                                                                                                                                                                 | Systems Programming    | Oct 2024 – Dec 2024 |
| <ul style="list-style-type: none"><li>Engineered 8-store virtual memory system in C for Xinu OS with Second-Chance replacement; reduced page faults by 25%.</li><li>Added 7+ kernel system calls, increasing memory performance by 15% using inverted page tables and a custom fault handler.</li></ul> |                        |                     |

Languages and Technologies

- Languages:** Python, TypeScript, Java, JavaScript, SQL, C, Bash, R
- AI/ML:** Anthropic Claude API, OpenAI API, Gemini API, LangChain, MCP, Hugging Face, TensorFlow, PyTorch, RAG
- Cloud & DevOps:** AWS, GCP, Docker, Kubernetes, Terraform, CI/CD pipelines, Grafana, Prometheus, OpenTelemetry
- Frameworks/Tools:** Node.js, NestJS, Next.js, Temporal.io, Kafka, FastAPI, Elasticsearch, Redis, PostgreSQL, Spark

Additional Experience and Awards

- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; developed 3 complete MVPs within 48-hour builds.
- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.